CLO-4: Bloom Taxonomy Level:<Understanding>

An employee at a software company received an email that looked like it was from the company's HR department. The email said there was an urgent update to the employee benefits policy and included a link labeled "View New Policy." When the employee clicked the link, it led to a login page that looked exactly like the company's internal portal. Without thinking, the employee entered their username and password. A few hours later, unauthorized activity was detected on the company's internal systems using that employee's credentials.

Question:

1. What type of security vulnerability is this?
2. Suggest one or two effective solutions to prevent this kind of attack.

CLO-4: Bloom Taxonomy Level:<Remembering>

A person named Faizan creates a fake website that looks exactly like the login page of a popular Pakistani bank. He then sends a mass email to thousands of people pretending to be the bank, claiming there's a security issue and asking them to log in using the provided link. Several people fall for the trick and enter their credentials. Faizan then uses these details to access their bank accounts, copy their private data, and attempt to transfer money. Later, it was discovered that Faizan had used a specially designed malware to capture usernames and passwords stored in plain text on infected computers.

Question:

Identify **at least three PECA laws** that Faizan has violated based on this scenario. Mention the section title and briefly justify your answer.

CLO-3: Bloom Taxonomy Level:<Applying>

Suppose you have received the following encrypted messages from the sender and you have been told that RSA algorithm is used to encrypt this information and the public key to decrypt the message is {7, 187}. Your task is to retrieve the original messages.

$C1 = 16$

$C2 = 24$

CLO-4: Bloom Taxonomy Level:<Applying>

1. Suppose you are writing an anti-virus software. You receive hundreds of samples of a single type of malware, but all samples look different. Identify kind of virus is that, and how can your anti-virus software spot it on running systems.
2. What are the 5 requirements to make a public-key crypto system a secure algorithm?